Group Participation Sheet

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A portion of this course is devoted to group work. This **Group Participation Sheet** (GPS), hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of "**zero**" for the mentioned assessment. Any concerns regarding this matter should be brought to the professor's attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

Clear Form Print Form

Course Title:	INFO-53883 - AI/ML Capstone Project	
Evaluation Name:	INFO-53883 - PR3 (5%, Group Work)	
Team Number:	INFO-53883 (AI/ML Capstone) - Team 2B	
Referencing and Citation Style:	*APA Style* ☐ *IEEE Style* ☐ *N/A* ☐	
Group Members:	*First Name and Family Name of group members:*	*Signature of group members:*
	(1) Afolabi Adesina (team leader)	Afolabi Adesina Digitally signed by Afolabi Adesina Date: 2026.06.22 11:46:38 -04'00'
	(2) Evans Frimpong	
	(3) Xian Qin	Xian Qin Digitally signed by 6d19637b-b2a2-4dbb-9e5e-df2e0d713851 Date: 2026.06.22 11:50:51 -04'00'
	(4) YANAN YANG	Yanan Yang Digitally signed by Yanan Yang Date: 2026.06.22 11:48:18 -04'00'
	(5)	
Submission Date:		

AI/ML Capstone Project
PR1 to PR7 Weekly Progress Report (5% of final mark)

Instructor: Prof. Mouhamed Abdulla

Name 1: Afolabi Adesina	**Student Number 1:** 991808102
Name 2: Evans Frimpong	**Student Number 2:** 991771602
Name 3: Xian Qin	**Student Number 3:** 991578381
Name 4: Yanan Yang	**Student Number 4:** 991841037

Q1: Summary of Work

Provide a summary of the work completed during this current sprint. [1 mark]

- **Yanan:** Trained three machine learning models, achieving accuracies around 82%. However, the F-1 scores were found to be extremely low for Class 2 (serious) and Class 3 (fatal). After conducting an investigation, it was confirmed that the dataset is heavily imbalanced: Class 1 (slight) accounts for 82.7%, while Class 2 and Class 3 account for only 15.6% and 1.6% respectively. This skew reflects real-life traffic accident distributions but poses a significant training challenge.
- **Evans:** Reviewed the application frontend framework and identified the critical API connections required to establish full functionality between the client side and the backend.
- **Xian:** Successfully test-ran Python scripts to fetch environmental variables via APIs from a government weather website. Additionally, discovered and reviewed a new dataset containing road condition images.
- **Afolabi:**
  - **Ethics & Fairness:** Investigated fairness metrics for the project to mitigate systemic bias in data collection.
  - **Literature Review:** Integrated two core research papers into the project framework: Paper 1 focusing on weather-aware trajectory planning and the SPI formula, and Paper 2 covering ML traffic prediction benchmarks.
  - **EDA & Schema Audit:** Explored both the City of Toronto and UK Department for Transport (DfT) datasets, conducted complete schema audits, and plotted feature distributions.
  - **Summary Statistics:** Replicated Paper 2's weather, road surface, and lighting factor analysis on local DfT data using custom bar charts and correlation heatmaps.
  - **Safety Score Design:** Calibrated individual weights for the E_{index} formulation using empirical paper findings (e.g., snow +47%, dusk +22%, standing water +21%).
  - **Preprocessing:** Handled missing values, engineered key temporal and environmental features (hour of day, day of week, is night, season), and executed label encoding.
  - **Correlation Analysis:** Computed Pearson correlation matrices, Chi-Square tests of independence, and Point-Biserial coefficients.
  - **Feature Selection:** Developed a three-method voting pipeline (combining Chi-Square, Mutual Information, and Random Forest Gini importance) to filter and select 8 optimal features.

  - **Model Training & Tuning:** Built a comprehensive model pipeline training 5 baseline models (Logistic Regression, Decision Tree, KNN, Random Forest, and LightGBM). Conducted hyperparameter optimization using GridSearchCV with StratifiedKFold cross-validation for Logistic Regression ($L1/L2$), Random Forest, and LightGBM.
  - **Model Comparison Matrix:** Generated a comprehensive performance evaluation table measuring Accuracy, Precision, Recall, Macro/Weighted F1-score, Matthews Correlation Coefficient (MCC), and Area Under the ROC Curve (AUC).

- **Additional Data for Images:** Incorporated external image repositories for vision-based classification:
  - Extreme Road Image Dataset: GitHub Repository
  - RSCD-1million Road Surface Classification Dataset: Hugging Face Dataset

Q2: List of Challenges

Provide a list of challenges encountered, and how the team addressed those challenges during this sprint. [1 mark]

- **Yanan:** The extreme class imbalance (82.7% Class 1) means that a naive model predicting the majority class automatically achieves high accuracy while failing completely on minority classes. Attempting to pass `class_weight='balanced'` within Random Forest and Logistic Regression alongside one-hot encoding caused the models to aggressively over-predict Class 2 and Class 3. This indicates that the current feature space may not linearly or easily separate the classes, demanding deeper non-linear feature engineering.
- **Evans & Team:** Encountered structural difficulties in securing a comprehensive, high-quality, and fully annotated image dataset suitable for training a robust computer vision model for automated road condition checking.
- **Xian:** Encountered accessibility constraints and strict language/protocol barriers with certain official government weather APIs. This required adapting connection scripts and looking into alternative open data feeds to ensure stable real-time data flows.

Q3: Technology Stack

Provide the current technology stack used with specific version numbers for tools, libraries, databases, and APIs. [1 mark]

- **Yanan:** Google Colab, `pandas`, `scikit-learn` (`neural_network`, `linear_model`, `metrics`, `compose`, `pipeline`).
- **Evans:** Flask API, YOLOv8 (Ultralytics architecture for image detection).
- **Xian:** Python environment, Apache Packages, `pandas`, `opendatatoronto` API wrapper.
- **Afolabi:** VS Code, Jupyter Notebook, Cursor IDE, PyTorch, TensorFlow, LightGBM, and Git.

Q4: Meeting Minutes

Provide the details of the meeting minutes (those conducted in class and outside) and the GitHub activity. [1 mark]

- **Yanan:** Summarized the analytical findings from the current model iterations and proposed a multi-experimental framework to enhance minority class F-1 metrics.
- **Xian:** Formulated algorithmic optimization strategies to enhance computational efficiency during data ingestion.
- **Afolabi:** Suggested integrating real-time highway camera feeds to build a 3-way decision fusion model (combining Tabular, NLP, and Vision brains).
- **Evans:** Conducted a detailed architectural assessment of state-of-the-art image recognition networks to focus development efforts on high-yield models.

Q5: Future Work

Provide a work plan for the upcoming week. [1 mark]

- **Yanan:** Design and run advanced modeling experiments to address dataset imbalance and evaluate feature separation viability.
- **Evans:** Establish a standardized output pipeline for the image training system and formulate the exact mathematical logic to integrate vision metrics into the final road safety score.
- **Afolabi:** Utilize Cursor to manage the pipeline architecture. Build the NLP Brain via a TF-IDF extraction pipeline on Ontario 511 text alerts, construct the Vision Brain using OpenCV and a Custom CNN on Ontario road camera images, and create a centralized multi-modal dashboard fusing all three safety components.
- **Xian:** Merge and map the newly acquired road image datasets with the existing database schema, and continue engineering real-time data flow pipelines from public transit APIs.

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